

Sanjay Vonk

<https://sanjayvonk.github.io>

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EDUCATION	Erasmus University Rotterdam, Rotterdam, The Netherlands <i>September 2021 – Present</i> <i>Bachelor Econometrics and Operations Research</i> Expected Graduation: September 2025 GPA: 3.7/4.0 (Upper Second-Class Honours Equivalent) Coursework: Real Analysis, Linear Algebra, Combinatorics, Advanced Probability (Stochastic Processes), (Multivariate) Statistics, Graph Theory, Optimisation (Linear, Non-Linear, Combinatorial), Time-Series Analysis, Applied Statistics, Simulation, Programming (Python, R, Matlab, Java, AIMS). Erasmus University Rotterdam, Rotterdam, The Netherlands <i>September 2021 – Present</i> <i>Bachelor Economics and Business Economics</i> Expected Graduation: September 2025 Major: Financial Economics GPA: 3.7/4.0 (Upper Second-Class Honours Equivalent) Coursework: Mathematical Auction Theory, Signaling and Moral Hazard, Financial Derivatives.
PUBLICATIONS	S. Vonk. (2024). <i>Conformalized Beliefs: Merging Distributional Modeling and Conformal Prediction</i>. In preparation.
MANUSCRIPTS	S. Vonk, A. Petric, M. Tejera Vicente, & T. Chang. (2024). <i>Enhancing Tumor Segmentation with Conformal Prediction</i>. Erasmus School of Economics. (Grade: 9.2/10) S. Vonk. (2024). <i>Enhancing Consumer Purchase and Brand Choice Modeling with Venn-Abers Calibration and Conformal Selection</i>. Erasmus School of Economics, <i>Best Research Paper (Grade: 9.1/10; Rank: Top 1/257)</i>. S. Vonk, B. Verhoef, R. Alexandrou, & D. Hulsegge. (2024). <i>Conformal Predictive Distribution Implied Moments and Stock Market Volatility Forecasting</i>. Erasmus School of Economics. <i>Best Research Paper (Grade: 9.9/10; Rank: Top 1-4/259)</i>.
PRESENTATIONS	<i>Enhancing Tumor Segmentation with Conformal Prediction</i> [Co-presented]. Turing Students Rotterdam, Turing Machine & Deep Learning, 2024 (TMDL-24).
RESEARCH EXPERIENCE	Conformalized Beliefs: Merging Distributional Modeling and Conformal Prediction 2024 Developing a theoretical paper that shows that conditional coverage can be achieved in conformal prediction by integrating prior distributional beliefs into the conformal framework. • Demonstrated how traditional conformal methods fit in the framework, and outlined theoretical conditions under which traditional conformal methods already achieve conditional coverage. • Provided practical methods for assessing and recalibrating incorrect beliefs, facilitating dynamic adjustments to better approximate conditional coverage without introducing new assumptions. • Extended the method to multivariate prediction tasks, addressing and overcoming the limitations of standard conformal methods when applied to high-dimensional data. Enhancing Tumor Segmentation with Conformal Prediction 2024 First authored a research project aimed at enhancing tumor segmentation models by incorporating Conformal Risk Control, to bound the False Positive Rate and significantly improving the interpretability and reliability of predictions in medical imaging applications. • Created per-pixel uncertainty estimates to segmentation models, offering insights into model confidence at each region of the classification, contributing to more reliable clinical decisions. • Developed an interactive 3D visualisation to highlight brain regions with high prediction uncertainty, providing visual aids for medical professionals working with brain scans in tumor diagnosis and treatment planning. [FEB22009Q] Enhancing Consumer Purchase and Brand Choice Modeling Through Venn-Abers Calibration and Conformal Selection 2024 Explored methods to improve probabilistic forecasting in quantitative marketing models. • Modelled consumer purchase behaviour and brand choice for six cereal brands in the USA. • Improved Black-Box model accuracy through Venn-Abers calibration and Conformal selection, with an emphasis on uncertainty quantification. • Controlled the False Positive Rate by introducing a withholding mechanism for uncertain predictions.

[FEB22009Q] Conformal Predictive Distribution Implied Moments and Stock Market Volatility Forecasting

2024

Conducted research on applying conformal predictive distributions to financial econometrics.

- Introduced a novel approach using Conformal Predictive Distributions to infer implied moments for stock market volatility forecasting.
- Demonstrated that CPD-Implied Moments achieve competitive predictive accuracy against established econometric models, particularly for short-term forecasting horizons.
- Presented empirical evidence that CPD-Implied Moments sometimes outperform traditional models, highlighting their utility in dynamic and unpredictable market conditions.

INDUSTRY
EXPERIENCE

Research Scientist

October 2024 – Present

MyArtBroker, London, United Kingdom (Remote)

- [REDACTED]
- [REDACTED]
- [REDACTED]

Quantitative Research Internship

March 2024 – October 2024

Velador Associates, London, United Kingdom (Remote)

- Developed theoretical frameworks for research and quantitative trading strategies.
- Conducted alpha research, resulting in the discovery of a successful trading signal.
- Demonstrated strong self-learning abilities and work ethic, leading to a full-time offer from MyArtBroker.

OTHER
EXPERIENCE

Head of Portfolio Management Education

August 2022 – Present

B&R Beurs, Erasmus Investment Society

- Designed and developed the curriculum for a 3-week Portfolio Management course for Europe's largest student-run investment society with over 1,850 members.
- Leading 4 instructors teach the course to 60 students, with weekly topics: Portfolio Theory, Market Efficiency, and Practical Considerations for Portfolio Management.
- Successfully passing students are awarded a full scholarship for the Chartered Alternative Investments Analyst (CAIA) Program, worth 2950 euro.

Lecturer in Financial Derivatives

August 2022 – Present

B&R Beurs, Erasmus Investment Society

- Teaching a class of 35 students on Credit Default Swaps and Collateralised Debt Obligations.
- Received offer to become Head of Derivatives Education after successfully redesigning lecture content.

LEADERSHIP

Chairman

August 2021 – 2023

Concordia International, International Student Investment Group

- Served as the 6th chairman of a co-educational society for students interested in finance, serving 25 active members and 30 alumni by organising two weekly meetings and annual alumni events.
- Competed every year in the B&R investment competition against ± 45 other investment groups, consistently ranking as best international investment group out of twenty, (top ten overall).
- Increased recruitment by obtaining 183 applications in the first year, strengthened alumni relations through events and trips, and improved education through organising lectures.

REFERENCES

Dr. Stuart Jamieson, PhD, MSc, CFA, FRM

Quantitative Researcher at Velador Associates

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Website: <https://research.ncl.ac.uk/nail/people/phdstuartjamieson.html>

Mentor for five years and supervisor during internship.

Dr. Eoghan O'Neill

Assistant Professor of Data Science and Machine Learning

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